Izar 2020 HGSOC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Both platforms and all 11 retained samples intermix on one connected manifold; the platform batch that sample-only Harmony leaves (same-platform frac 1.00) is removed by joint platform+sample Harmony (→0.65). No islands.

Gate 2 — module geography → GO (non-branching)

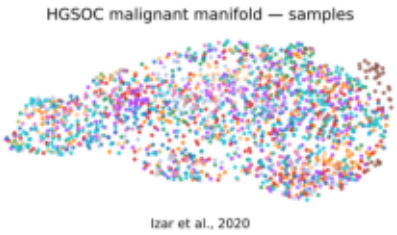
Shared Gast/pSom gradient (geography r=+0.53; reproduced within each platform) with Crest the inverse, terminal end (Crest~Gast −0.68, Crest~pSom −0.92; Crest-high decile at pt 0.81).

Gate 3 — cross-repression → NA

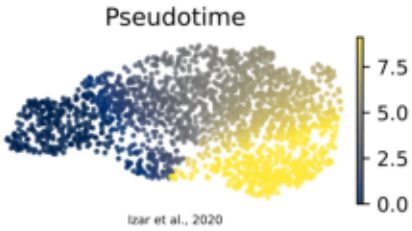
Non-branching principal curve (0 forks across Nodes 15–30), so there is no hub to test.

STEP FIGURES

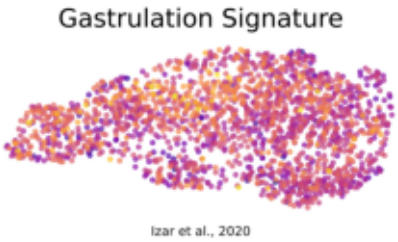
Manifold — samples



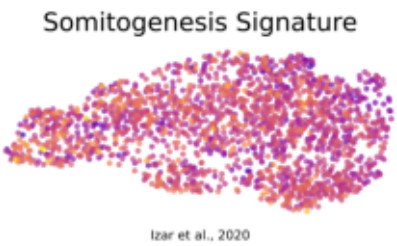
Pseudotime



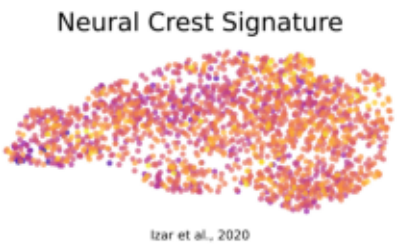
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC malignant ascites (Izar et al., 2020), two platforms (SmartSeq2 1,158 + 10x 751 malignant); 1,909 loaded → 1,904 after thresholding. Both leaves concatenated on the shared gene space; platform treated as a batch covariate alongside sample.

Verdict.
The embryonic-program pattern reproduces as a single non-branching axis — a Gast/pSom-high developmental origin flowing to a neural-crest-high terminus, with proliferation orthogonal to (not aligned with) the axis.
Consistent with the Linear-pSom/Crest outcome seen across most of the atlas.

Parameter choices

Min cells/sample	10 (mult-gap 21→5; keeps 11/12)	**Integration**	Harmony [platform, sample] θ=10; PCA20
Scope	whole manifold (no lobe scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip Gast/pSom-high, Crest-low	**Geometry**	Linear (non-branching); 0 forks
EMT-TFs	SNAI2 (root) → PRRX2 (terminus)		